

ASHUTOSH SHARMA

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Generative AI Engineer | Data Scientist | RAG & LLM Systems

PROFESSIONAL SUMMARY

Generative AI Engineer with 3.5+ years building production RAG platforms, agentic LLM systems, and ML pipelines. Progressed from ML analytics to leading enterprise Agentic AI development — architecting multi-node workflows that lifted answer faithfulness 40% to 80%+ and cut operational cost 50%. Stack: LlamaIndex, FastAPI, PostgreSQL/pgvector, Celery, Kubernetes.

KEY HIGHLIGHTS

- **80%+ answer faithfulness** on enterprise RAG via NLI Citation Verification + CRAG filtering, cutting hallucinations 60%.
- **6x ingestion format coverage** — added JSON, XLSX, CSV, DOCX, and HTML parsers on top of PDF baseline.
- **Sub-second p95 latency** on streaming RAG service over 5,000+ daily requests on FastAPI + Kubernetes.
- **92% tool-selection accuracy** on ReAct Agent across 6 specialized retrieval tools with loop prevention.

TECHNICAL SKILLS

Generative AI & LLM: LlamaIndex, LangChain, CrewAI, RAG, Agentic AI, ReAct Agents, Query Decomposition, Prompt Engineering, RAGAS, CRAG, NLI Citation Verification, OpenAI API, Llama, Hugging Face, Transformers, BERT

AI/ML: NLP, Deep Learning, Topic Modeling, Classification, Clustering, Feature Engineering, scikit-learn, PyTorch, LightGBM, Random Forest, XGBoost, Pandas, NumPy, Matplotlib, Seaborn

Backend & Data: Python, FastAPI, Async Programming, PostgreSQL, SQL, pgvector, Hybrid Search, Semantic Search, SQLAlchemy, Celery, RabbitMQ, WebSocket, REST APIs, Streaming APIs, Vector Databases

Cloud, DevOps & MLOps: Docker, Kubernetes, Helm, Git, CI/CD, MLOps, Prometheus, Loguru, Phoenix Tracing, AWS, Azure, Object Storage (S3), Linux, Jupyter Notebook

Tools & Methodologies: Confluence, Jira, Agile, Scrum, TDD, Code Review, System Design

Soft Skills: Communication, Teamwork, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Associate Consultant | T-Systems ICT India Pvt. Ltd.

Dec 2022 – Present

Bangalore, India

AI Operator – Agentic RAG Platform

May 2025 – Present

- Architected multi-node RAG platform on FastAPI, LlamaIndex, and PostgreSQL/pgvector, reducing query response time 35% under production load.
- Designed Query Decomposition, ReAct Search, Synthesis, and Citation nodes as composable LlamaIndex Workflow steps.
- Engineered ReAct Agent across 6 specialized retrieval tools, achieving 92% tool-selection accuracy with loop-prevention safeguards.
- Integrated NLI-based Citation Verification and CRAG filtering, raising faithfulness from 40% to 80%+ and cutting hallucinations 60%.
- Extended ingestion pipeline to support JSON, XLSX, CSV, DOCX, and HTML, onboarding 15+ enterprise data sources.

- Reduced manual preprocessing 65% through programmatic schema normalization and structured table extraction for pgvector.
- Built RAGAS evaluation harness with Phoenix tracing across 1,000+ weekly runs, enabling 15% faster prompt-iteration cycles.
- Deployed on Kubernetes with Helm, maintaining 99.5% uptime via Prometheus monitoring and memory-bounded Celery workers.

CrewAI Multi-Agent Content System (POC)

Dec 2024 – Apr 2025

- Orchestrated modular CrewAI multi-agent framework for scalable content generation across 5 workflow use cases.
- Cut content production time 80% on blog creation pipeline while maintaining editorial quality standards.

Insurance Healthcare Agent – Agentic RAG System

Jul 2024 – Nov 2024

- Developed agentic RAG system unifying unstructured documents and PostgreSQL into a single retrieval layer.
- Improved query-resolution efficiency 40% and customer satisfaction 35% across the support workflow.
- Optimized LLM-powered agent workflows, slashing agent workload 50% and customer wait times 70%.
- Launched production service handling 10,000+ monthly queries at 98% accuracy and 99.5% service availability.

Topic Modeling for Document Clustering

Feb 2024 – Jul 2024

- Analyzed 1,000+ enterprise documents using LDA topic modeling, boosting searchability and content organization 45%.
- Trained LightGBM classifier reaching 90% accuracy on extracted topics, reducing manual tagging effort 70%.

Enterprise Data Analytics & ML Modeling

Dec 2022 – Jan 2024

- Designed customer segmentation and classification models using XGBoost and Random Forest, achieving 87% prediction accuracy.
- Streamlined ETL pipelines processing 500K+ records with feature engineering, reducing data preparation time 60%.
- Delivered interactive analytics dashboards and reporting workflows for enterprise clients, improving decision cycles 30%.

EDUCATION

Master of Science in Data Science | Manipal Academy of Higher Education, Manipal, India | GPA: 8.9/10

Jul 2022 – Jul 2024

Bachelor of Science in Computer Science, Mathematics & Statistics | Bangalore North University, Bangalore, India

Jul 2018 – Jul 2021

CERTIFICATIONS

- **Generative AI with Large Language Models** – DeepLearning.AI / Coursera
- **LangChain for LLM Application Development** – DeepLearning.AI
- **Machine Learning Specialization** – Coursera (Andrew Ng)

LANGUAGES

English – Professional Working Proficiency | Hindi – Native